

# Thais Isabelle Parron Ruiz

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## EDUCATION

University of South Florida - Tampa, FL  
Bachelor of Science in Mechanical Engineering  
Minor: Aerospace Engineering

GPA: 3.87  
Expected Graduation Date: Dec 2026

## EXPERIENCE

### Lead Tutor / Engineering Tutor

AUG 2024 – Present

University of South Florida

Tampa, FL

- Promoted to Lead Tutor in August 2026 to coordinate engineering tutoring center, including schedules, attendance, schedule-change announcements, and weekly student sign-in records.
- Tutor students in Statics, Dynamics, Thermodynamics, Electrical Systems I, Programming, Computational Methods, Vibrations, and Kinematics & Dynamics of Machinery.

### Vibration-Based Corrosion Detection Research

MAY 2026 – Present

University of South Florida

Tampa, FL

- Perform impact hammer testing on uncorroded and corroded steel plates using accelerometers to measure and compare changes in natural frequencies, frequency response functions, mode shapes, and nodal patterns.
- Contribute to MATLAB code development for a vibration-based method that compares experimental and numerical modal data to identify the most likely corrosion location.

### PFAS-Free Fog Harvesting Research

AUG 2024 – DEC 2025

University of South Florida

Tampa, FL

- Designed and tested a PFAS-free fog harvesting system using laser-textured aluminum surfaces with superhydrophobic and patterned superhydrophilic/superhydrophobic regions to improve droplet capture, transport, and collection.
- Integrated Peltier cooling and evaluated untreated, superhydrophobic, and patterned samples under controlled fog conditions, with the patterned surface achieving the highest collection at **6.884 mL**.

## ENGINEERING PROJECTS

### Dual-Action Can Crusher Design

JAN 2025 – MAY 2025

Kinematics & Dynamics of Machinery Final Project

- Designed a dual-action can crusher in SolidWorks using a **6-bar** linkage mechanism to pre-dent cans horizontally before completing vertical crushing.
- Reduced required input force by approximately **40%**, from **36** to **21** pounds, through prototype testing with mechanical and electrical force gauges.

### Ladder Logic Control System Project

JAN 2026 – MAY 2026

Mechanical Controls | LogixPro PLC Simulator

- Developed PLC ladder logic programs in LogixPro for door control, two-way traffic control, and conveyor/silo automation using relays, timers, counters, sensors, and output coils.
- Tested automated logic for motor control, traffic light timing, proximity-based box counting, reset conditions, interlocks, and status indicator lights.

### Computational Methods with MATLAB

JAN 2025 – MAY 2025

Computational Methods

- Applied MATLAB to solve engineering problems in fluid mechanics and heat transfer using numerical methods, matrix systems, finite differences, regression, interpolation, and spline fitting.
- Compared theoretical and numerical results using plots, relative error analysis, and engineering assumptions such as laminar flow, round-off error, and heat conduction behavior.

### Projectile Simulation Game

MAR 2024 – APR 2024

Programming Concepts

- Developed a MATLAB projectile motion simulator using user-defined launch height, target location, launch angle, velocity, and gravity conditions for Earth, Mars, and Jupiter.
- Created dynamic trajectory visuals, input validation, multiple attempts, and feedback for maximum height and distance from the target within a **2-meter** accuracy requirement.

### Acoustic Guitar CAD Model

AUG 2023 – DEC 2023

Computer Aided Design and Engineering

- Modeled a full acoustic guitar assembly in SolidWorks from scratch, including detailed part drawings, material selection, exploded views, and manufacturing-style documentation.
- Designed a functional tuning mechanism using shaft-mounted worms and worm gears to adjust string tension through rotational motion.

## SKILLS

**Software:** SOLIDWORKS, MATLAB, Simulink, LabVIEW, LogixPro PLC Simulator, Autodesk Fusion 360, Microsoft Office

**Programming/Tools:** Arduino, MATLAB scripting, PLC ladder logic, CAD modeling, 3D printing

**Languages:** English (fluent), Portuguese (native), Spanish (basic).

**Certifications:** SOLIDWORKS Associate, Mechanical Design and Additive Manufacturing